Introductie
Experimental Research Project

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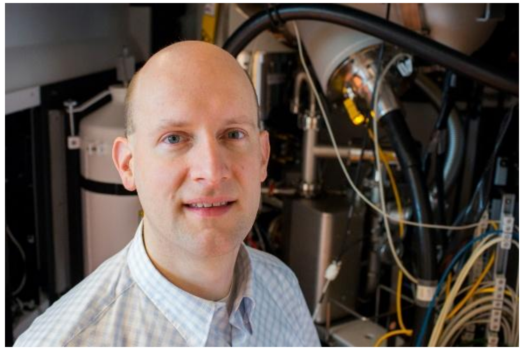
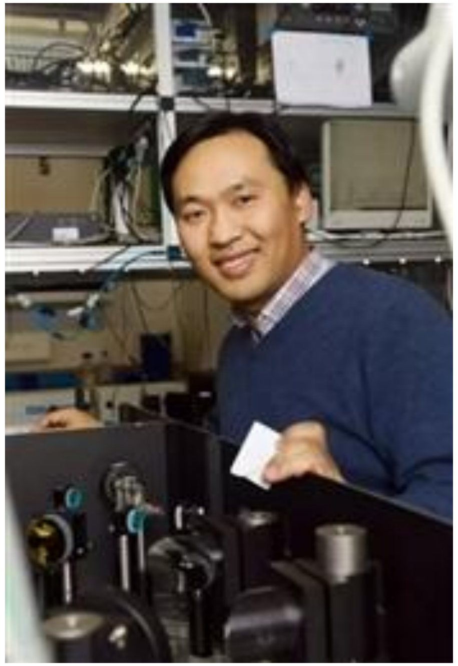




Course outline

- Waarom de cursus ERP?

Observations

Slide courtesy of Peter van Capel

- Compared to other Dutch universities

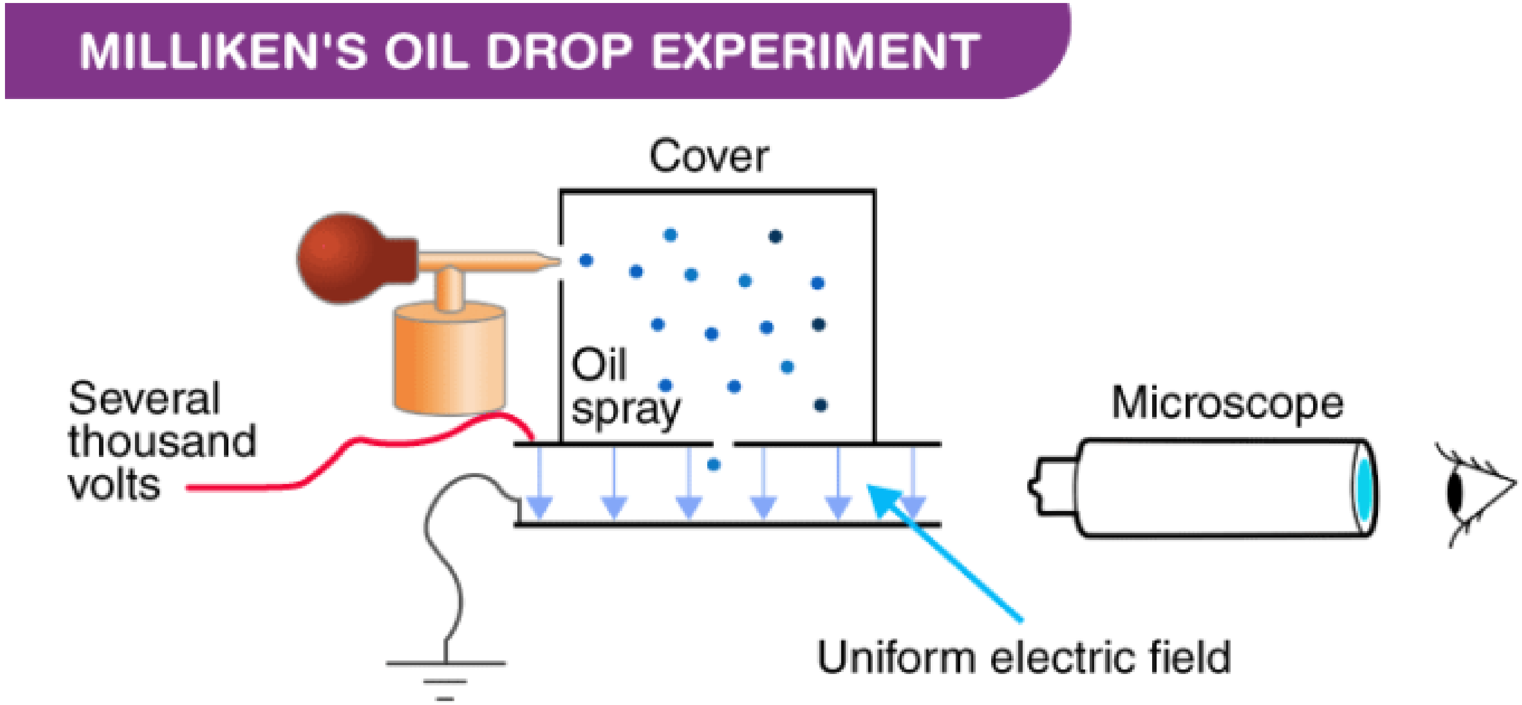
University	Mandatory	Elective
UU	8 ec	7.5 ec
UL UvA/VU RUG RU	11 ec 16 ec 20 ec 15 ec	6 ec 6 ec - 9 ec (max)
TUD TUE UT	30 ec 25 ec 23 ec	5 ec

- Main differences are in 2nd / early 3rd year on advanced level

[**Overview lab courses at Dutch / Flemish universities**]

Outline

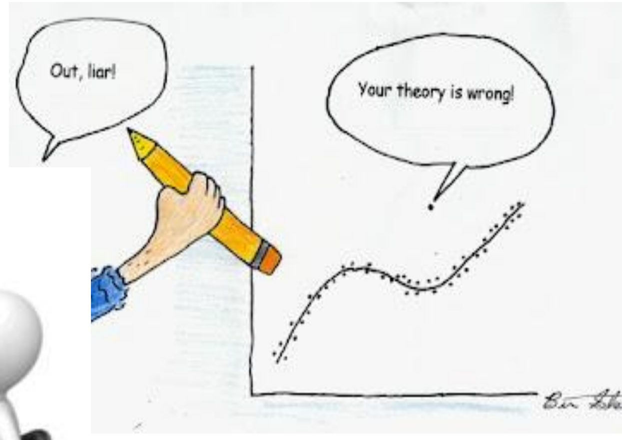
- Waarom de cursus ERP?
- Vrijere opdrachten, meer nadruk op ontwerpen experiment



Targets: going though a research cycle

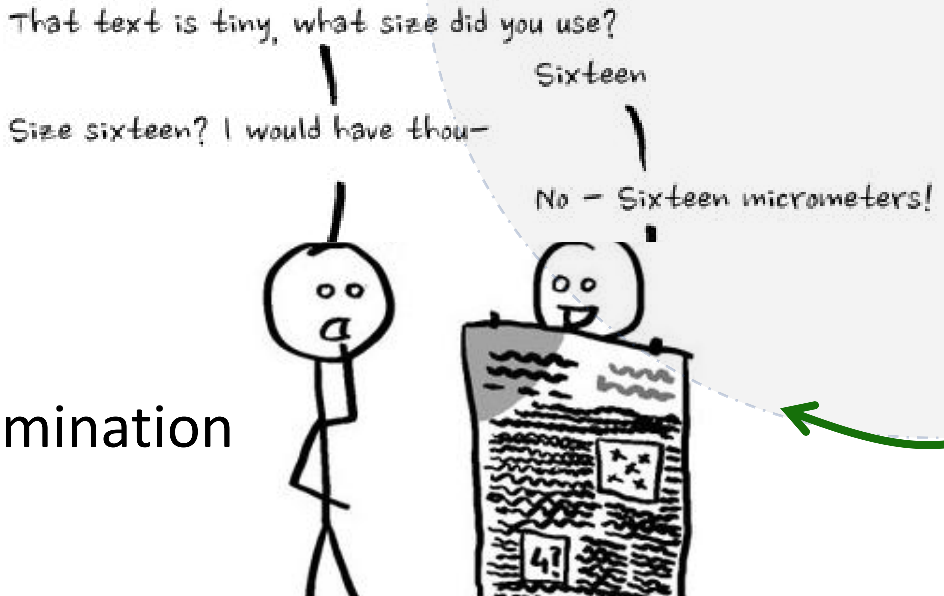
Question

Implementation: setting up, measurments...



Interpretation: analysis & modelling

Dissemination

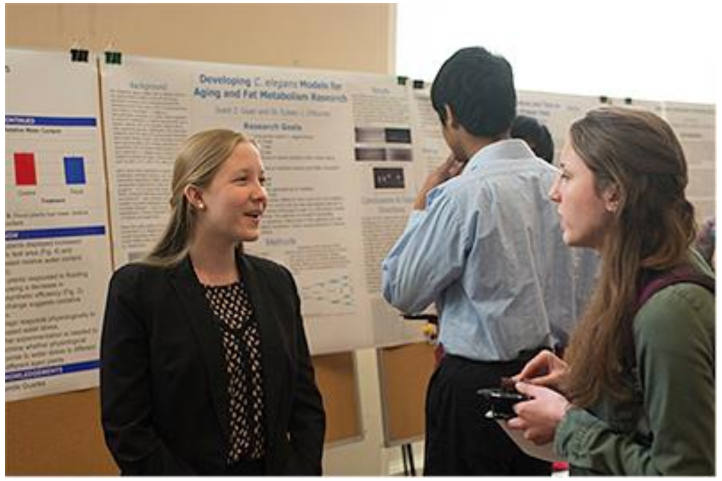


Outline

- Waarom de cursus ERP?
- Vrijere opdrachten, meer nadruk op ontwerpen experiment

Inhoud van de cursus:

- Lectures – Intro en samenwerken — week 1
- Modules – Training vaardigheden — week 2
- Project — week 3-9
- Poster presentatie (24.06; 13-16:00) — week 10
- Reflectie op project — week 10



ERP All-in-One Document

Welcome.

This all-in-one document includes all information for teachers, students, and teaching assistants (TAs) for experimental research project, 2026. Please send an email to Hai Wang (h.wang5@uu.nl) and cc Marijn van Huis (M.A.vanHuis@uu.nl) if something is unclear or missing. We will keep the documents updated while do our best to keep the structure logical and clear.

ERP All-in-One Document

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The information on Osiris seems outdate (which we will try to solve asap)

Please use **Brightspace** and the course document there

1. Introduction to the course and schedule

2. Learning goals and objects

3. Implementation: When/where who to do what -guides for students/teacher/TAs

   3.1 Guides to students

   3.2 Guides to TAs

   3.3 Guides to teachers:

4. Information for general lectures and Modules in week 1-2

5. Project description (for week 3-9)

   5.1 A quick overview on the teacher and project (to be updated for 2026)

   5.2 Project description

6. Group Project Proposal Handout

7. Rubric for evaluation

Group formation (Week0)

Student-ID	Naam	tussenv	achternaam	Opleiding	Voorziening	group	first
				NAST		1	yes
				NAST		1	
				NAST		1	
				NAST / WISK		2	yes
				NAST / WISK		2	
				NAST		2	
				NAST / SCHK		3	yes
				NAST / SCHK		3	
				NAST / SCHK		3	
				NAST / WISK		4	yes
				NAST / WISK		4	
				NAST / WISK		4	
				NAST		5	yes
				NAST		5	
				NAST		5	
				INCA / NAST		6	yes
				NAST		6	
				NAST		6	
				NAST		7	yes

Under course Information/BS

Week 1: Lectures

Lecture 1
22 april

- Project market: introduction to research projects (9:00-12:00; MIN 001-004)

Lecture 2
22 april

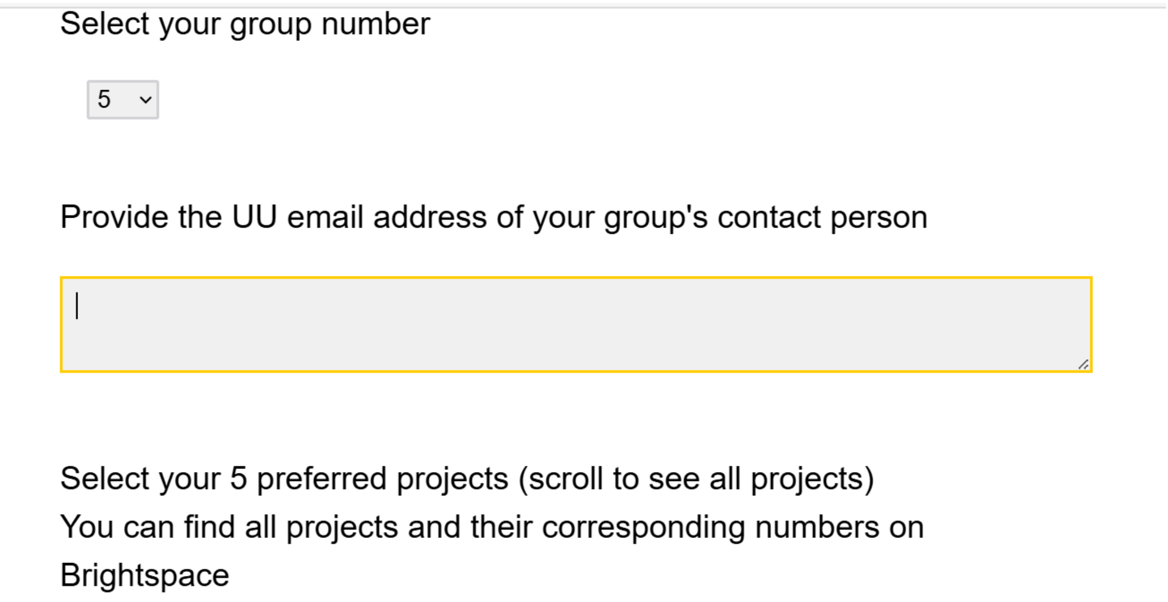
- Introduction to the course; Q & A
(13:15-15:00; KGB Atlas)

Lecture 3
24 april

- Collaboration and collaboration grid (related to an assignment with the deadline on 15/05/2026)

Week 1: submitting your project wishes

Submitting 5 project wishes to: https://survey.uu.nl/jfe/form/SV_8oi9BjnUtmAc70q


Select your group number

5

Provide the UU email address of your group's contact person

Select your 5 preferred projects (scroll to see all projects)
You can find all projects and their corresponding numbers on Brightspace


First choice

Second choice

Third choice

Fourth choice

Fifth choice

1

2

3

4

5

By 26/04/2026, 23:59:59

Project assignment (beginning of week 2)

ject number	Teachers	Project name
1	Erik van Sebille, Marc Schneiter	Measuring water surface movements with a self-built floating GNSS tracking system
2	Erik van Sebille, Marc Schneiter	Data collection and analysis for an outdoor GNSS tracking system
3	Maurice van Tiggelen	Meteorological measurements 1: Weather station
4	Maurice van Tiggelen	Meteorological measurements 2: solar UV/VIS/IR radiation and wind measurements
5	Rupert Holzinger & Nemat Omidikia	Environmental Research Using Sensitive and Specific Mass Spectrometer (I)
6	Rupert Holzinger & Nemat Omidikia	Environmental Research Using Sensitive and Specific Mass Spectrometer (II)
7	Bibhasvata Dasgupta, Jacoline van Es, Alexis Gilbert, Thomas Röckmann	Isotopic composition of methane (CH_4) emitted from different sources
8	Getachew Adnew, Alexis Gilbert, Thomas Röckmann	Performing and interpreting H2O isotopologue signals
9	Allard Mosk	The electro-optics cookbook
10	Allard Mosk	I spy with my little lensless camera
11	Dries van Oosten	Acoustic experiments
12	Dries van Oosten	Open project or The physics of musical instruments
13	Peter van der Straten	Speed of Light
14	Peter van der Straten	Saturation spectroscopy
15	Hai Wang	Who am I? or open project
16	Marta Verweij	Muon Tomography
17	Thomas Peitzmann	Muon lifetime
18	Alessandro Grelli	Muon Tracking with GEMs
19	Anuradha Samajdar, Chris van den Broeck,	Thermal Expansion with an Interferometer
20	Raimond Snellings	Open project

The project list, numbers, and detailed descriptions can be found under Course Information on BS.

Project assignment (beginning of week 2)

Matching results: 28/04/2026 on Brightspace

https://github.com/mverwe/ERP-matching

Import favorites | Google Calendar - E... | rud | Goodgame Big Farm | Goodgame Big Farm

README | GPL-3.0 license

`process_CSV_qualtrics.cc` or `process_CSV_qualtrics.py`

Matching projects to students

To avoid biases we first randomize the order in which groups are considered with `randomize_top_picks.cc`. Next we will run the matching code (`matchmaking.cc`) many times (for example 100 thousand) to make sure we sample a large phase space of possible configurations. The matching code performs the following steps:

1. Generate a random number, A, to select a project for the group being considered (p_i is the probability to get the ith preferred project and is an adjustable parameter of the code)

- $0 < A < p_1$ → top 1 selected
- $p_1 \leq A < (p_1 + p_2)$ → top 2 selected
- $(p_1 + p_2) \leq A < (p_1 + p_2 + p_3)$ → top 3 selected

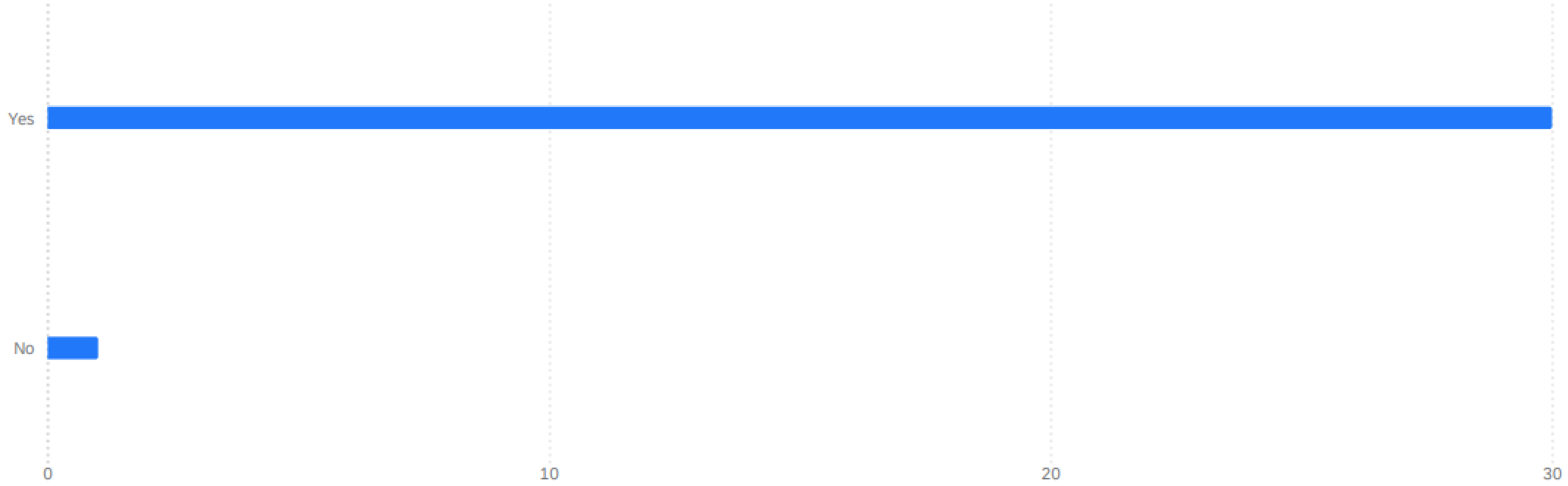
2. Is the project already assigned to another group?

- Yes: repeat step 1
- No: assign project to group

3. If no project from top 3 is available, select a project randomly
4. Calculate happiness factor of the student grouop

Maximize the "average happiness" by a code

It is OK to NOT get your dream project



- Half of the teams did not get their first choices;
- They are satisfied eventually with the project that they worked on

Week 2: Modules (29/04-01/05, slot A and D)

week 2

We will be assigned to one of the modules in week 2:

1. Electronics and signal processing
2. Optics and spectroscopy (max. 24 plaatsen)
3. Statistical data analysis
4. Mass spectrometry
5. Prototyping at Lily's Proto Lab (max. 24 plaatsen)
6. Data-acquisition and sensors (ARDUINO)

For each project, the teacher has indicated which modules the three students will be assigned to. You don't need to choose a module
On 28/04, we will upload your group's project assignment results, as well as the module assigned to each individual student in Brightspace (lecture room/time will be shared as well then).

Project implementation (week 3-9)

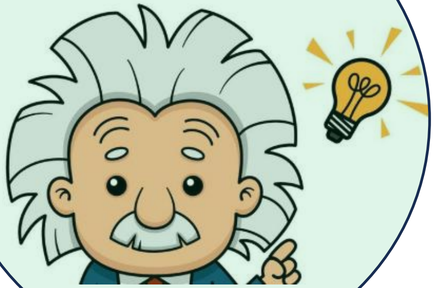
Time	Task	Comments
Week 2	Contact your project teacher by an email	
Week 3-9	Design, build, measure, document; Weekly progress meeting;	
Deadline 1: 15/05/2026	Assignment: submitting your proposal of max 3 pages to BS	Including (1) project aims; (2) how will you solve the problem; (3) How will you work together

You are the owner of your project!

Supervision in the project phase (Week 4-8)

You are the owner of your project!

PIs



TAs serve as intermediate contacts to help project implementation

They are discussion partners, helping students to find a way to resolve issues (not directly provide solutions)

- Weekly supervision meeting with teachers

Students

Technicians

9 TAs/4ad vanced TAs (ATAs)

Project implementation (week 10)

Time	Task	Comments
Week 2	Contact your project teacher by an email	
Week 3-9	Design, build, measure, document; Weekly progress meeting;	
Deadline 1: 15/05/2026	Assignment: submitting your proposal of max 3 pages to BS	Including (1) project aims; (2) how will you solve the problem; (3) How will you work together
Deadline 2: Week 10, by 22/06 noon, 12:00 am	Assignment: submitting your poster to BS	With a suggested format provided by the coordinator
Week 10, 24/06/2026 13-16:00	Poster session	4 poster prizes (2 for designs + 2 for content)
Week 10, 25-27/2026	Final feedback meeting with TA and teachers	5/8/26

2 mandatory group assignments!

Lab Journal

A lab journal should record all experimental details, raw data, observations, and interpretations in enough detail for another researcher to reproduce the work.

It also serves as an official legal record of the research. Essential components include the date, title, objectives, detailed procedures, materials used, numerical and qualitative data, sources of error, analysis, and conclusions.

1 Workplan

Date	What we plan to do
Week 1 (12 to 16 May)	Part 1: walk through the article and discuss, start making Workplan. Part 2: Finish workplan, visualize the setup and get an understanding of the setup.
Week 2 (19 to 23 May)	Part 1: exploring the setup and divide the creation of the setup into 'stages'. Start working with the laser. Start with creating the first stages of the setup. Also work on the background of the experiment for the poster. Part 2: continue building the setup.
Week 3 (26 to 30 May)	Part 1: continue building the setup while simultaneously testing the setup. Part 2: finish building the setup. Don't forget to take pictures of the setup. Also start (test) measuring.
Week 4 (2 to 6 June)	Part 1: finish building the setup with the audio frequency. Part 2: finish the whole setup. Continue writing setup and method and start working on the poster.
Week 5 (9 to 13 June)	Part 1: Check alignment one last time and start measuring. Continue with poster and start thinking about presentation. Also finish measuring sinc this isn't expected to take long. Part 2: Analyse the data of the measurements

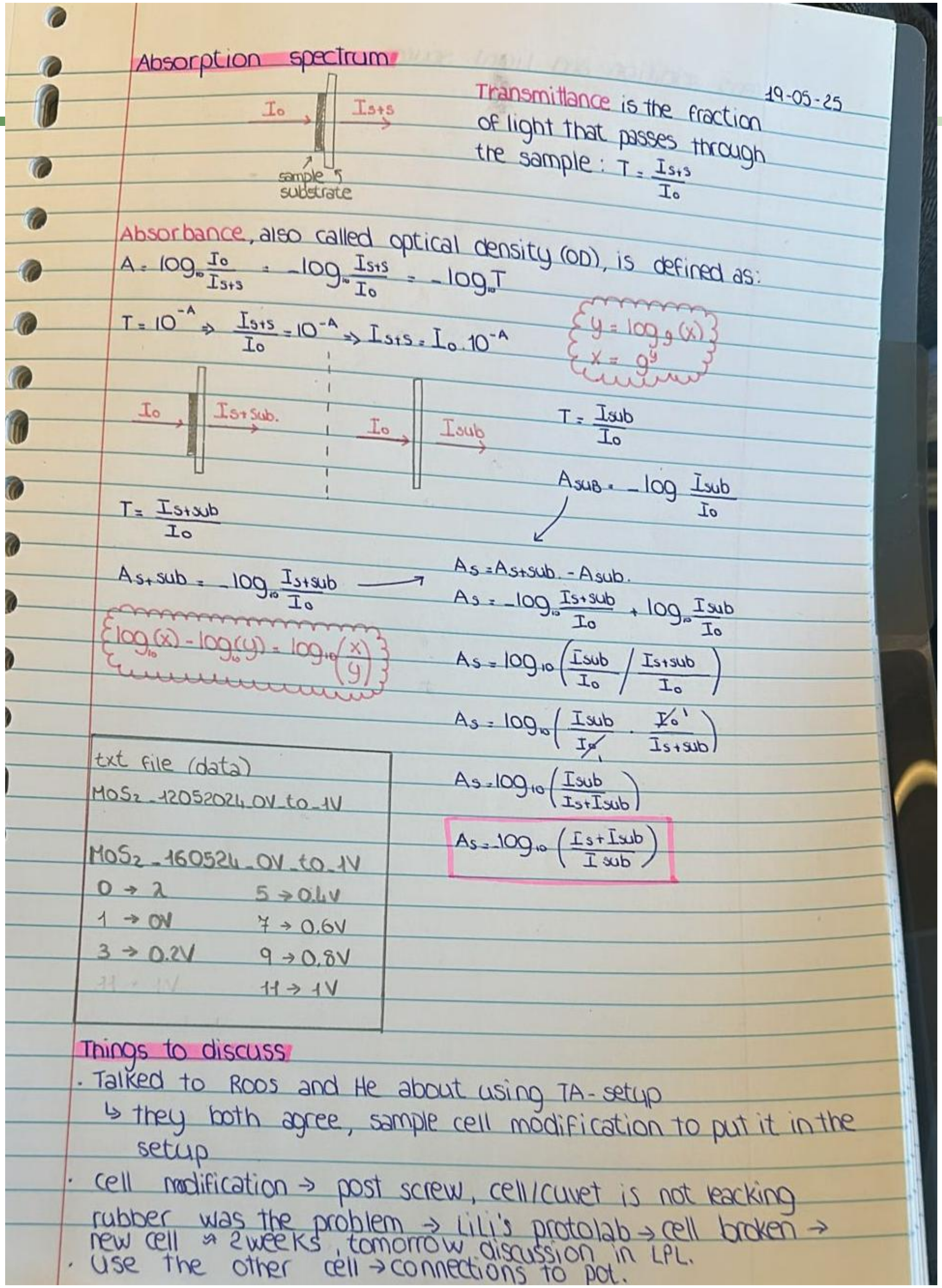
Date	**What we did**
Monday 12 May	We saw the setup and talked with Hai about the project.
Tuesday 13 May	We talked the article over and discussed what we needed to do. We asked the questions we wanted to ask and we got started on the Workplan.
Friday 16 May	We started working on the LaTeX and the Workplan. We also dove deeper into the theory of the experiment and we got started on learning the setup. Professors helped us build and understand the setup, which was very nice. We finished our workplan and talked through the next stages of the experiment.
Tuesday 20 May	Today we started by dividing the building of the setup into different stages to make it more clear. After this we started working with the laser and tried to connect the photodiode to the oscilloscope. However, we missed a cable and couldn't do this. We hope to continue this tomorrow.
Wednesday 21 May	Today we continued exploring the setup. We once again tested the laser and now we wanted to try to connect the photodiode and create a signal on the oscilloscope, since now we have the right cable. Unfortunately we only saw noise on the apparatus. We didn't discover what went wrong and we also asked the TA and other professors. In the meantime we continued writing the introduction and the background for the poster. Unfortunately today the problem isn't yet solved, but we hope to solve it this Friday. We finished the introduction and divided the background into parts so we can start writing it easily.
Friday 23 May	This morning we started off with the help of Jan Bonne. We couldn't see anything other than noise and he explained to us why this is. From this we worked with Jan Bonne and Rudy to make the preparations for the rails. Then we attached the laser to the rails and the photodiode too. We worked a long time on aligning the laser and the photodiode. Eventually this worked so we could start modulating the optical carrier with the RF signal and hope we get a signal out of this. Unfortunately we also ran into some problems here and it took us a while to adjust. Now we have a new setup and on Monday we can finish building that and we can start adding the audio signal.
Monday 26 May	Today we start with scheduling a meeting with Peter to see what he thinks of our progress. After this we continued coming up with ideas for the setup. Last

	know we obtained a reasonable result with test measuring.
Friday 13 June	Today we will start with some preparations so after this we can start measuring after which we can write the results and the outlook. We also discussed some good things to know for the presentation and enhancement of the poster with our TA Lotte. We ran into some other problems and we found out that our frequency mixer had blown up. We got a new one and now we can finally see the signal again. We measured three times in total and we also had one test measure so in total we measured four times. Now we will start on drawing conclusions from our results. We divided the tasks for the poster between us and we will work hard on this this weekend.

Table 1: Logbook

3 Introduction

From Empedocles to Hertz, and from Galileo to 17th CGPM, we have been fascinated by the speed of light since Ancient Greece. The determination of this constant was a big breakthrough and is since its discovery still one of the most important fundamental constants. From this universal constant comes the exact definition of the meter: the path traveled by the speed of light in a certain time interval. This speed defines daily concepts, but is also a crucial part of many theories and experiments. An accessible and not overly complex way of measuring the speed of light is not only vital for non-physicists to be able



Absorption spectrum

19-05-25

$I_0 \rightarrow$ | $\rightarrow I_{s+s}$

sample, substrate

Transmittance is the fraction of light that passes through the sample: $T = \frac{I_{s+s}}{I_0}$

Absorbance, also called optical density (OD), is defined as:

$$A = \log_{10}\frac{I_0}{I_{s+s}} = -\log_{10}\frac{I_{s+s}}{I_0} = -\log_{10} T$$

$$T = 10^{-A} \Rightarrow \frac{I_{s+s}}{I_0} = 10^{-A} \Rightarrow I_{s+s} = I_0 \cdot 10^{-A}$$

$y = \log_g(x)$
$x = g^y$

$I_0 \rightarrow$ | $\rightarrow I_{s+sub.}$ $I_0 \rightarrow$ | $\rightarrow I_{sub}$

$T = \frac{I_{sub}}{I_0}$

$A_{SUB} = -\log\frac{I_{sub}}{I_0}$

$T = \frac{I_{s+sub}}{I_0}$

$A_{s+sub} = -\log_{10}\frac{I_{s+sub}}{I_0} \longrightarrow$

$\log_{10}(x) - \log_{10}(y) = \log_{10}\left(\frac{x}{y}\right)$

$A_S = A_{s+sub} - A_{sub}$

$A_S = -\log_{10}\frac{I_{s+sub}}{I_0} + \log_{10}\frac{I_{sub}}{I_0}$

$A_S = \log_{10}\left(\frac{I_{sub}}{I_0} \Big/ \frac{I_{s+sub}}{I_0}\right)$

$A_S = \log_{10}\left(\frac{I_{sub}}{\cancel{I_0}} \cdot \frac{\cancel{I_0}}{I_{s+sub}}\right)$

$A_S = \log_{10}\left(\frac{I_{sub}}{I_{s+I_{sub}}}\right)$

$\boxed{A_S = -\log_{10}\left(\frac{I_{s+I_{sub}}}{I_{sub}}\right)}$

txt file (data)	
MoS_2 _12052024_0V_to_1V	
MoS_2 _160524_0V_to_1V	
0 → λ	5 → 0.4V
1 → 0V	7 → 0.6V
3 → 0.2V	9 → 0.8V
~~11 → 1V~~	11 → 1V

Things to discuss

- Talked to Roos and He about using TA-setup
  ↳ they both agree, sample cell modification to put it in the setup
- cell modification → post screw, cell/cuvet is not leacking rubber was the problem → Lili's protolab → cell broken → new cell ≈ 2weeks, tomorrow discussion in LPL.
- use the other cell → connections to pot.

Implementation

TASK	ASSIGNED TO	START	END
Group formation (weak 0)	Students	4/8/25	4/16/25
Project assignments (week 1-2)	students/Course coordinator	4/23/25	4/28/25
Elective module (week 2)	students/Course coordinator	4/29/25	5/8/25
Start project (weak 4)	Student groups	5/8/25	5/8/25
Design, build, measure, log (weak 4-8)	Student groups	5/13/25	6/13/25
Weekly progress meeting (weak 4-8)	Students + PI+ TAs	5/8/25	6/13/25
Presentation of the results			
Prepare poster (weak 4-9)	Students	6/5/25	6/17/25
Present poster	Students	**6/18/25**	6/18/25
Final feedback meeting (weak 10)	Students + PIs + TAs + technicians	**6/23/25**	6/27/25

The purpose of a poster

1. A communication medium to help you reporting your finding to public

The purpose of a poster

1. A communication medium to help you reporting your finding to public

2. Networking tool

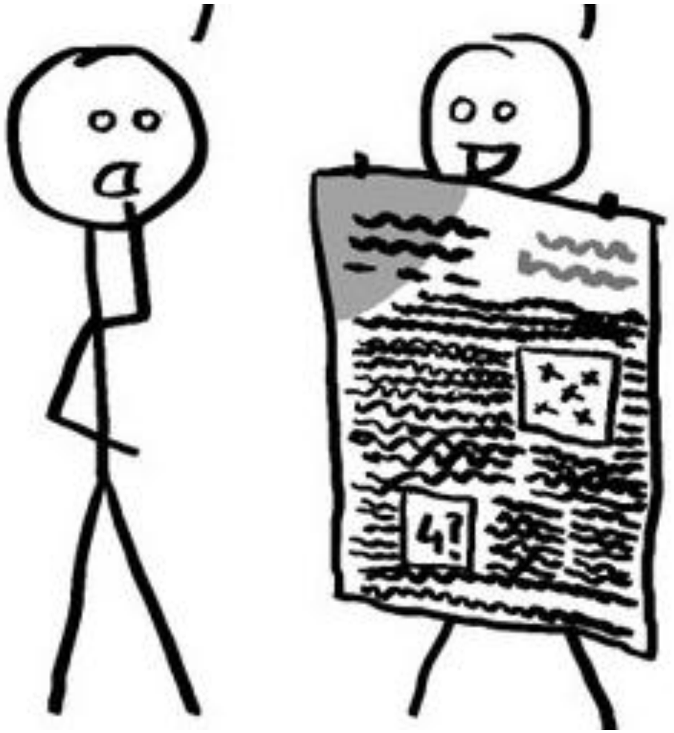
What a poster is NOT for

- The test is so tiny.
What size did you use?

- sixteen

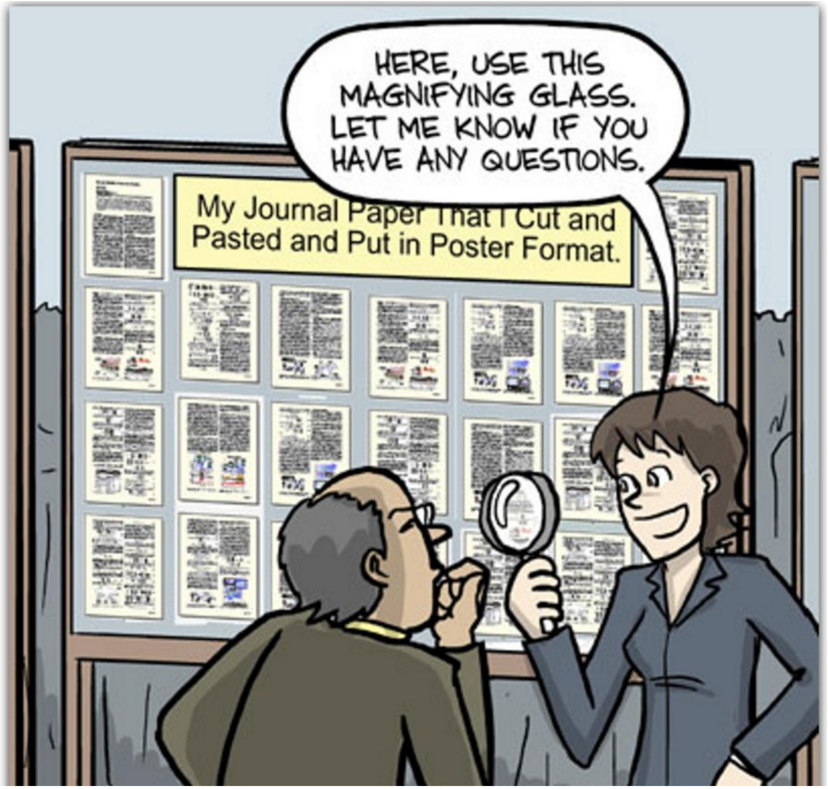
- size sixteen? I would have
thought....

- no, sixteen micrometers



Information bombardment
i.e. too wordy

“Kill your darlings”



Tell one coherent story, with only necessary, per-selected data!

A good poster may cover the following aspects

- Background + Research questions: Why and what?

- Methodology: How?

- Result and discussions. What/Why?

- Conclusions and outlook: where to go?

- Bonus: Include a catchy illustration or image that highlights the key physics, finding, or product from your project.

We will suggest one possible design, but you are free to create yours

HOW TO DESIGN AN AWARD-WINNING CONFERENCE POSTER

Dr. Tullio Rossi

ANIMATE YOUR SCIENCE
www.animate-science.com

#1 SCRIPTING

- YES to bullet points - NO to long paragraphs.
- Use sections with HEADERS.
- Maximum 250 words! Possibly <150.
- Don't forget your contact information.
- Make sure your poster is telling a story that includes:

Background | Question | Methods | Results | Conclusions

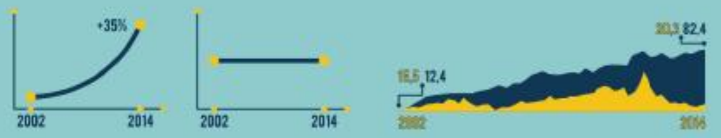
#2 DESIGN

- Decide a layout before you start designing.
- Negative space is your friend. 40% should be blank.
- Use 3 to 5 colors.
- Use 1 accent color to draw attention.
- NO to images and patterns as background.
- Use 1 to 2 fonts - readable from 1 m.
- Feel: More like an infographic less like a scientific poster.

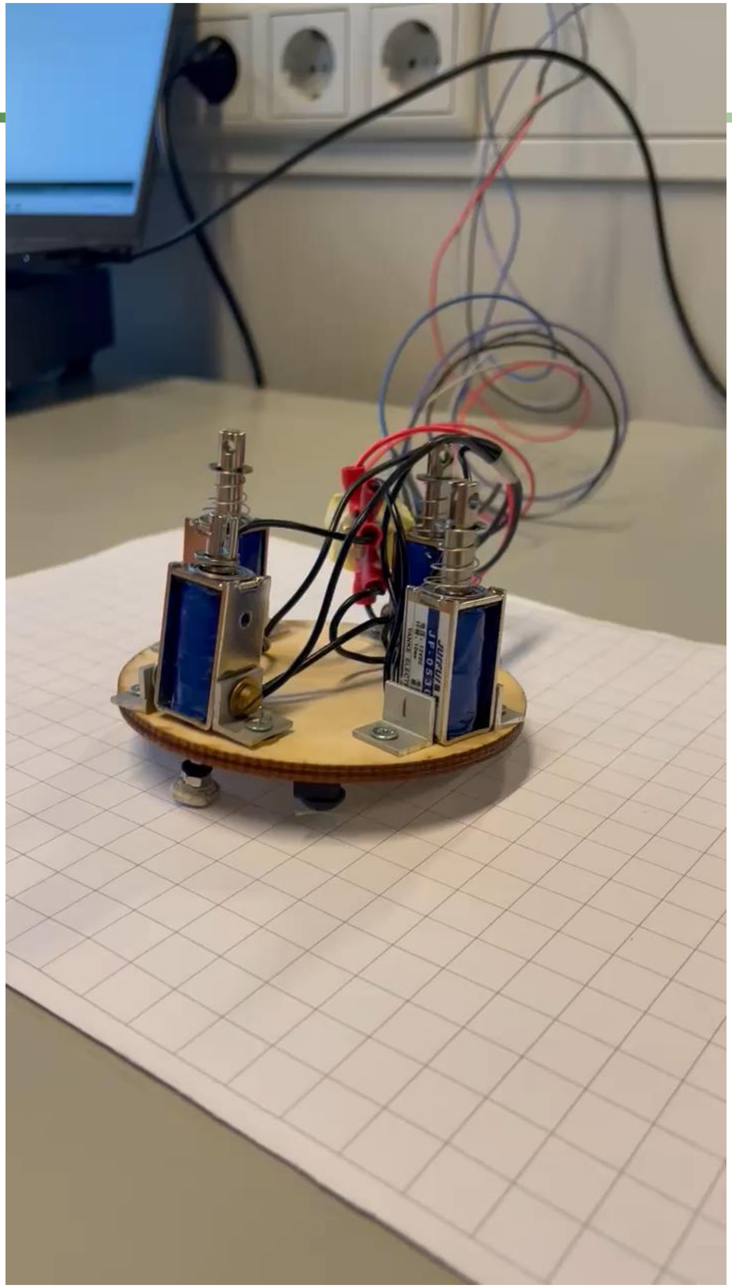
Include one large eye-grabbing visual

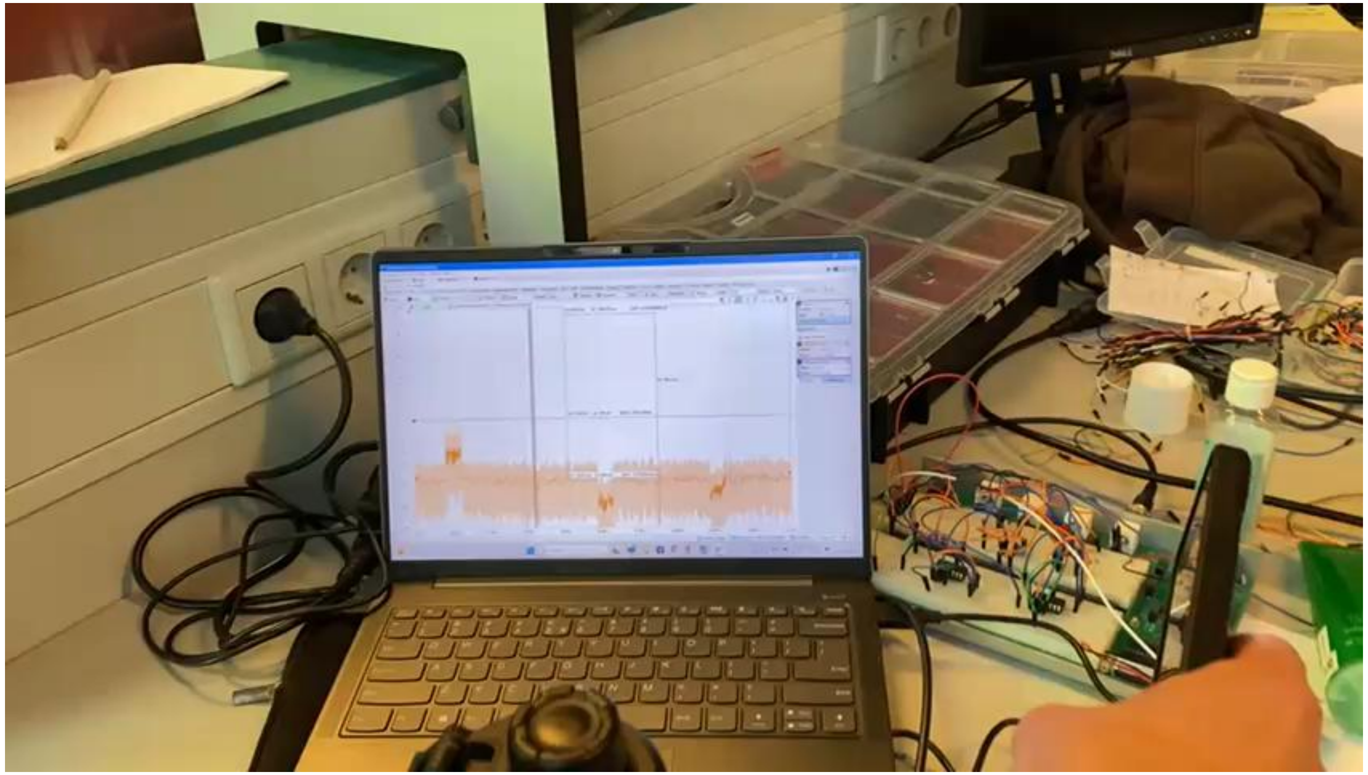
#3 DATA

- Display only the essential.
- Simplify graphs to make them easier to read.
- Apply the color scheme to the graphs for consistency.



Poster session was a big success





Vragen?